

# Structural review of the housing–fertility OLG model: consolidated reading of two referee reports

Table 5.1 (repository-reading) and ChatGPT Pro, same master prompt

September 15, 2026

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## How to read this document

Two reviewers answered docs/prompts/MASTER\_PROMPT\_model\_review\_20260915.md on September 15, 2026. Fable read the repository and verified every code claim against file and line (docs/model/structural\_model\_review\_fable\_20260915.md, commit 3f6b5f0b). ChatGPT Pro worked from the prompt, the slides and the ledger text (docs/model/structural\_model\_review\_chatgpt\_20260915.md, verbatim copy). For each of the 25 labels this file gives: what the two agree on, condensed; where they disagree, both positions; what only one of them raised, attributed; and the two recommendations side by side. Neither review is edited here; where a reviewer’s claim was checkable against code or a local PDF, the check is recorded. Notation follows the prompt:  $n$  children ever born,  $m$  children at home,  $\bar{h}(m)$  the parents’ space floor,  $e(m)$  the equivalence scale,  $\kappa_H$  the housing taste scale,  $\kappa_1, \kappa_C$  the fertility taste scales,  $\xi$  the first-birth cost.

## Where the two reviews stand

**Both retain the mechanism and the household block.** Both say the core (down payment at purchase, owner-only large units, children raising the value of space, priced fertility timing) is coherent and should not be rebuilt, and both locate the real faults in accounting, closure, supply and finance rather than in preferences. Both say the same three things loudest: the child maturation process is memoryless and must change; the housing stock must have an inherited-stock transition; the population, person and dependent accounts must be written as one contract before any recalibration.

**Both independently found the same error in the prompt.** The prompt’s description of the population closure (imposed historical masses, headship law after 2023) describes an older branch, not the retained one, which propagates households endogenously with the births-over-2.1 queue.

**The split is about how much structure to add.** ChatGPT is conservative: fix descriptions, normalizations and accounts first; add economic structure (amortization, concave child utility, a child time cost, a rental wedge) only when a specific behavioral failure demands it. Fable is prescriptive: where the reference papers or the data already contradict an implementation choice, change it now (stock-flow supply, an amortizing origination-only mortgage, a size-dependent rental wedge in place of the cap and the owner premium, one coherent earnings decomposition, a measured child earnings penalty in place of  $\xi$ ). The disagreements below are all instances of this split.

**Sequencing differs.** ChatGPT: units and demographic closure first, then preference and credit interpretations, then earnings and geography, then identification. Fable: the two parameter-level repairs first (matura-

tion, earnings coherence), then the household block (mortgage, tenure shock) in one refit, then supply with the terminal condition, then geography with the final refit.

**Raised by only one reviewer.** ChatGPT: welfare with endogenous fertility needs an explicitly identified population; the down-payment test excludes four years of current earnings; the at-most-one-birth-per-period restriction; the level normalization of a negative bequest utility under child scaling; net depreciation versus paid maintenance; do not reapply the  $\psi_0$  root in counterfactuals; the scale in Scholz–Seshadri–Khitatrakun multiplies utility (verified, and it changes the cost of children from falling to rising in  $m$ ). Fable: no time cost of children, hence no negative income–fertility gradient; estates have no receiver; no widowhood or downsizing shock; two supply elasticities coexist in code; live depreciation is 1.1 percent, not 2; the tenure-shock value flips the policy’s ownership sign;  $h_P$  is at its upper bound; one price per room at every size; the household-formation margin is frozen.

## Preferences

### P1. Utility form and the linear child term

**Agreed.** The building blocks (CRRA over goods and space net of a parents’ floor, an equivalence scale, a reduced-form benefit of children at home) are defensible, and the paper must write their rationale. The linear benefit  $\psi m$  gives no diminishing marginal benefit, and at  $\sigma = 2$  with scale exponent 0.7 the flow utility of an additional child at fixed goods and space is convex among households already parenting: the scale burden rises less than proportionally (Fable’s increments 0.234, 0.216, 0.203 for the first three children). Both stress that this does not prove corner completed fertility; it places the stopping margin on dynamics, constraints and the taste shocks rather than on preferences.

**Disagreement: what to do.** ChatGPT: keep provisionally. Describe the object as parental utility over equivalized living standards plus the benefit of children at home, and do not introduce concave child benefits or a different cardinal scaling until a specific behavioral failure justifies it. Fable: author decision. Position 1, keep linear and state that  $\kappa_C$ , the fecundity decline and the window carry the intensive margin; position 2, a concave  $\psi v(m)$  with externally fixed curvature (Sommer 2016 and Barro–Becker put curvature in the child term), which makes the intensive margin an economic object. Discriminating observable: number of children among mothers by household income and the three-plus share.

**Only ChatGPT, verified.** Scholz, Seshadri and Khitatrakun’s objective is  $E \sum_j \beta^{j-S} n_j U(c_j/n_j)$  with  $n_j = (A_j + 0.7K_j)^{0.7}$  (their pp. 615 and 619, checked in the local PDF): the scale multiplies utility as well as deflating consumption. The model uses  $U(c/e)$  only, so it imports the scale’s shape, not their objective, and with endogenous family size the difference matters for behavior and welfare. Do not multiply by  $e$  merely to match the citation; that would be a substantive change. Becker–Lewis does not derive a linear  $\psi m$ .

*Consolidator’s note on what the verified form implies.* Under the SSK objective at  $\sigma = 2$  the flow utility is  $-e(m)^2/X$ , so the cost of successive children is 0.522, 0.580, 0.630 in units of  $1/X$ , rising in  $m$ , whereas under the model’s  $U(c/e)$  it is 0.234, 0.216, 0.203, falling. Adopting the cited paper’s actual objective therefore turns the convexity that both reviewers flag into a diminishing net benefit of children without any change to the child term, at the price of a larger first-child cost and a recalibration of  $\psi, \xi, \kappa_1, \kappa_C$ . This is a third option for the author decision below, not a recommendation from either review.

**Only Fable.** Doepke and Kindermann (2019) are the linear precedent: a one-time linear utility from each birth, paired with a fixed goods cost, a fixed utility cost and a wage-scaled time cost per child, so their marginal cost of children is not concave. The scale exponents 0.7 and 0.7 are hard-coded literals (solver.

py:2278);  $\psi$  multiplies children at home (solver.py:2254);  $h_P = 2.3$  sits at its upper bound with the per-child rooms slope restricted to zero.

**Recommendations.** Fable: Decide (low; blocks). ChatGPT: Keep (low; blocks, the cardinal interpretation must be chosen).

## P2. Sign and role of the cross-partial

**Agreed, fully, on the mathematics.** The prompt’s worry is reversed: with  $u = \mathcal{C}^{1-\sigma}/(1-\sigma)$ ,  $u_{sm} = \mathcal{C}^{-\sigma}(\mathcal{C}_{sm} - \sigma \mathcal{C}_s \mathcal{C}_m / \mathcal{C})$ , so with  $\mathcal{C}_m < 0$  the CRRA curvature strengthens a positive cross-partial (ChatGPT’s derivation). For the implemented aggregator with a parenthood-only floor,  $\mathcal{C}_{sm} < 0$  for  $m \geq 1$  because the scale lowers  $\mathcal{C}_s$ , while  $u_{sm} = (\sigma - 1)(e'/e)u_s > 0$  at  $\sigma = 2$  (both). The slide states the restriction on  $\mathcal{C}$ ; it must be restated on  $u$ , with finite differences because children are discrete (both; ChatGPT cites latex/september\_14\_presentation.tex:127–145, 414–428). The marginal rate of substitution  $u_s/u_c = \frac{1-\alpha}{\alpha} \frac{c}{s-h(m)}$  does not depend on  $e(m)$  (both): additional children do not shift the intratemporal space–goods trade-off; only the first child does, through the floor; larger families occupy larger homes only through income, saving and selection.

**Only Fable.** Both channels raise  $u_s$  only when  $\sigma \geq 1$ ; at  $\sigma < 1$  the scale channel reverses. At fixed expenditure the floor moves rooms by  $\alpha \Delta h$ : 1.69 rooms for an unconstrained renter against 1.01 realized, the gap being the constraints (cap, sale cost). The scale multiplies future marginal utility by 1.23 for a first child, which is where anticipatory saving for a down payment comes from.

**Only ChatGPT.** A positive  $u_{sm}$  does not establish the sign of the optimized fertility response to prices; three properties must be kept apart (marginal utility of space, relative housing demand, fertility response to cost). Add a per-child space component only if the paper intends and supports a separate requirement per additional child.

**Recommendations.** Fable: Keep (none; no). ChatGPT: Change (low; no). Same substance: restate the property on  $u$ .

## P3. The first-birth cost $\xi$

**Agreed.**  $\xi$  shifts the first-birth index by  $\pi_a \xi$ ;  $\kappa_1$  rescales that index and its response to costs; the two are not interchangeable, and  $\xi$  is a childlessness lever by construction, which is fair rather than an objection. The identification mapping on the slide is a reading of the parameter table, not a verified Jacobian (M25).

**Disagreement: keep or replace.** *ChatGPT:* keep, as an explicit fixed utility cost of entering parenthood, not an unspecified goods expenditure. Run the zero- $\xi$  restriction holding the  $\psi_0$  root; removal may preserve the childlessness mean but there is no guarantee it preserves the joint age–wealth pattern of first births. *Fable:* author decision, lean change. Replace it with a measured per-period earnings cost of children at home (a child penalty on  $y^d$  while  $m > 0$ , Kleven et al.), which scales with the wage and yields the negative income gradient of fertility; keep  $\xi$  only if childlessness still needs it. The nearest precedent, Doepke and Kindermann’s fixed utility cost per child, sits beside a time cost and cannot be borrowed alone.

**Observables.** Both: first-birth hazards by age and liquid wealth together with completed childlessness. Fable adds childlessness by permanent income or education, on which  $\xi$  predicts a flat gradient and an earnings penalty a negative one.

**Recommendations.** Fable: Decide, lean change (low; blocks). ChatGPT: Keep (low; no).

## P4. Children ever born and children at home

**Agreed.** Keep both. A household with  $m = 0$  may be childless or an empty nest, and the two face different first-birth costs, scales and remaining opportunities; the three ever-born moments need  $n$ . Fable quantifies why the confusion is not rare: under memoryless maturation 22 percent of first children leave within four years.

**Implementation notes.** ChatGPT: store only feasible combinations  $0 \leq m \leq n$  and share continuation computations outside fertile ages. Fable: drop  $n$  from the state after the last fertile age, since nothing after 46 depends on it once the bequest is child-blind.

**Recommendations.** Keep (both; low or none; no).

## P5. Bequests

**Agreed.** The child-blind warm glow is the right object (De Nardi; Kopczuk and Lupton); do not restore scaling in the number of children; the estate must be net bequeathable wealth at a consistent death-date price, net of mortgage and any liquidation cost. Fable verified that the code values the house gross of the 6 percent selling cost (`solver.py:2455–2459` against `solver.py:2442`), so dying in the house is cheaper than selling it.

**Only ChatGPT.** For  $\sigma > 1$  the unnormalized  $B$  is negative, so multiplying it by the number of children injects a fertility incentive through the level normalization; child-specific altruism would need recipients and an estate-division rule. De Nardi’s framework does not imply a preference for bequeathing housing over financial assets; old-house retention must come from actual frictions.

**Only Fable.** The estate has no receiver: it is a utility argument and the wealth leaves the economy, a choice Greaney et al. also make for their entrants. For a paper about intergenerational allocation, route estates to households aged 45–65 as a lump sum; this also relieves the wealth-to-earnings undershoot that pushes  $\beta$  to its cap. The larger omission for the policy is the absence of every downsizing force (widowhood, care, health; Venti and Wise 2004, to verify): the old release large homes only at death, so a holding tax moves them only along the intensive margin at a 6 percent cost. Author decision on an age-rising forced-move shock.

**Recommendations.** Fable: Change the two implementation points (medium; blocks). ChatGPT: Keep the net-estate motive (low; blocks, the net-estate accounting must be settled). Substance close; the receiver question is Fable’s alone.

## Fertility and children

### F1. Timing within the period

**Agreed.** Keep. The sequence (fertility shock, attempt, conception, housing shock, housing choice) is an information structure, not a solution device: households adjust housing within the four years in which a birth occurs, which is the object the PSID rooms response measures. Committing to a house before conception is a different problem and belongs in a targeted robustness run, not in the baseline. Do not read the sequence as households waiting until delivery to move.

**Only Fable.** The September 6 nested-choice experiment found the joint-versus-sequential difference at fixed continuation values to be of order  $10^{-6}$  births per household; it did not test commitment.

**Only ChatGPT.** Observables should include moves before unsuccessful or postponed fertility plans where they can be seen.

**Recommendations.** Keep (both; none or low; no).

## F2. The two extreme-value shocks

**Agreed.** The fertility shocks are economic primitives (unobserved motivations, identified by the level and age spread of first births). The housing shock at  $\kappa_H = 0.005$  is a search-bound artifact with no dedicated moment, not an estimate, and it should leave the estimated vector. The extreme-value location normalization must be stated: Fable notes the code's  $\kappa \log \sum \exp$  is exact only for mean-zero shocks; ChatGPT notes that omitted scale-dependent constants are not harmless when scales differ by reproductive history.

**Disagreement: what replaces the estimated  $\kappa_H$ .** *ChatGPT:* fix it as a numerical approximation with a stated limiting target as smoothing vanishes; check menu dependence, since  $K$  equal-valued alternatives add  $\kappa_H \log K$  to the log-sum and grid refinement must not create a taste benefit. *Fable:* either  $\kappa_H = 0$  with the kink handled at the distribution level, or estimate it against a tenure-transition moment (the PSID four-year Brier score dropped on July 24). It cannot stay a free numerical choice: the August 20 vintage at  $\kappa_H = 0$  gives a positive ownership response to the tax and the September 1 vintage at 0.010 a negative one. Greaney et al. choose tenure and size deterministically, so a product-level taste shock is not “DUE-style”.

**Recommendations.** Change (both; Fable low to medium, ChatGPT low; blocks).

## F3. Stochastic maturation, deterministic adult ageing

**Agreed.** The asymmetry is fine; memorylessness is the fault. With per-period exit  $\mu$ , duration has no minimum and no maximum: newborns enter the first draw and dependents persist into parental old age (Fable:  $\mu = 4/18$ , 22 percent of newborns gone within four years, 8 percent of children born at parental age 25 still home at 65). Independence across siblings is secondary. If the process stays, ChatGPT would rename it “departure from dependency” and stop calling exits maturation.

**Disagreement: the fix.** *ChatGPT:* a bounded stochastic dependency duration through a short child-age-stage or birth-history state. Stochasticity can remain, memorylessness need not. This reopens the author's recorded decision to set bounded durations aside and is, in ChatGPT's words, the largest substantive household-state decision in the package. *Fable:* no new state. Let the exit hazard depend on the parent's age,  $\mu(a)$ , low through the fertile window and rising to one by 62; since the last birth is at 46 and mortality starts at 66, every child leaves before any parent can die. Add a one-bit newborn flag that exempts a newborn from the first draw. An Erlang-2 law with the same mean cuts newborn exits to 7 percent but leaves 25 percent at home after 24 years, so a two-stage clock alone does not fix the tail. *Trade-off between the two:* the parent-age hazard zeroes the orphan flow and thins the tail at zero cost in state space, but it mis-times departures for children born early in the window, who stay until the parent's hazard rises; the child-stage state gets timing right at the cost of state space and the reopened decision.

**Observable (both).** Dependents at home by parent age and time since birth (ACS), which the model overstates above 60 and understates at 25–35.

**Recommendations.** Change (both; Fable low to medium, ChatGPT medium; blocks).

## F4. Dependents of a dying household

**Agreed.** If F3 removes dependency before any parental mortality, the problem disappears and no care pool is needed; that invariant should be established rather than a repair added. Joint death changes the mortality

model rather than the bookkeeping; reassignment adds state and fertility effects.

**Disagreement, conditional.** If the current dependency process stays, ChatGPT prefers an aggregate care pool (goods and housing needs, funding, an exit rule, no second entry stream) over joint death or reassignment; Fable treats all three as symptom repairs and would build none.

**Only ChatGPT.** The 1.2 percent flow was measured on an earlier checkpoint, not the retained baseline; and because the recorded-birth queue already preserves those children’s future entry, the established gap is their care and housing, not a loss of persons.

**Only Fable.** Verified code: `run_e5f_open_population_transition.py:847–851, 887`; the 2023 measurement is 1.01 percent of dependents.

**Recommendations.** Change (both), through F3.

## F5. Exogenous fecundity, chosen attempts

**Agreed.** Keep the split (Sommer 2016’s absorbing infertility shock; Doepke–Kindermann’s birth probability with natural fecundity and imperfect birth control; de la Croix–Pommeret’s fitted schedule).  $\pi_a$  must be success conditional on the modeled attempt, not an observed birth rate that already contains choices.

**Only ChatGPT.** The restriction of at most one birth per four-year period imposes spacing and limits catch-up after delayed first births; losses in completed fertility must not all be attributed to biology. Attempts are costless, so estimated attempt probabilities at very low  $\pi_a$  are not literal. Define the four-year attempt and map biological evidence to that horizon.

**Only Fable.** The live schedule (0.02, 0.134) gives 0.97, 0.90, 0.83, 0.71, 0.50 at ages 22, 30, 34, 38, 42, within three points of Léridon’s four-year figures, but was chosen by eye. Every birth is chosen, whereas about 45 percent of U.S. pregnancies were unintended in 2011; an exogenous birth hazard for non-attempters (author decision) would lower the price elasticity of total births by roughly the intended share.

**Recommendations.** Keep (both). Fable: low, does not block. ChatGPT: medium, blocks (horizon and birth-count mapping).

## Housing and finance

### H1. The constraint set

**Agreed.** The current formulation lets an owner re-borrow up to the collateral limit each period against an unchanged house, so “no refinancing” is not literally true (Fable: `kernels.py:979–985`, at no cost). The slide’s down-payment inequality is wrong for stayers: its right-hand side is zero, leaving a net-equity restriction, whereas the code tests only on trades (ChatGPT’s reading; Fable: `kernels.py:640–662`). The gross-return factors are not an error; housing trades precede current earnings by design.  $\phi$  stays.

**Disagreement: how much to add.** *ChatGPT:* change the formal description and the purchase-condition logic only. State the down payment as a restriction on purchase transactions; specify incumbent debt and new borrowing separately; describe the signed-asset formulation as a reduced-form collateral technology. Do not add rate risk, amortization and HELOCs at once; preserve the parsimonious core. *Fable:* add a required amortization share, apply the collateral test at origination only, forbid cash-out for stayers, and add a payment-to-income test if the Coven comparison is to be like-for-like. Free cash-out mutes Coven’s cash-flow channel (the old pay the holding tax out of equity) and removes the buffer-stock motive of young

owners. Verified: Greaney et al. have the model’s current block (origination LTV, no amortization); Kaplan, Mitman and Violante and Coven et al. both have amortization and payment-to-income limits.

**Only ChatGPT.** The down-payment test excludes four years of current earnings from purchase liquidity. At a four-year period this is a strong assumption and, in ChatGPT’s view, the more immediate concern.

**Recommendations.** Change (both; medium; blocks), with different content.

## H2. The underwater rollover

**Agreed.** Grandfathering existing debt after a price fall is right; Kaplan, Mitman and Violante distinguish origination limits from limits on existing mortgages. The age taper is ad hoc, must be exposed in the paper, and must not be deleted from the code just to match the slides.

**ChatGPT.** Replace or rationalize the taper with a contract rule that distinguishes scheduled outstanding debt from new borrowing; inherited negative equity is not permission to originate unsecured credit; specify the creditor and settlement rule for sale shortfalls and terminal unpaid debt.

**Fable.** The taper disappears under H1’s amortization; otherwise disclose it as an assumption. Code notes: one taper array doubles as the unsecured-credit scale (inert at  $\lambda_d = 0$ ); sale shortfalls carry into the renter state and decay under the same taper (`solver.py:3471–3474`).

**Recommendations.** Change (both; Fable low, ChatGPT medium; blocks).

## H3. The rental cap

**Agreed.** The cap is an approximation of segmentation, not an institutional fact. Greaney et al. cap rental size and calibrate the level from the size distribution (Fable verified: 5.39 against 10.78, the 90th percentile of rentals over the 10th percentile of owner units), so “6 rooms, Greaney et al.” should become that recipe measured in the AHS. The paper needs a soft-tail alternative (size-specific rental supply or a steep size-dependent premium) and the joint size–tenure distribution to discipline it; the question is whether the result needs literally zero access or survives expensive, limited access.

**Disagreement: the baseline.** *ChatGPT:* a hard-cap baseline can remain, with the soft-tail comparison as robustness. *Fable:* make the size-dependent wedge the baseline,  $r_t(h^R) = r_t + w(h^R)$ , calibrated to AHS renter shares by size; the wedge is Kaplan, Mitman and Violante’s landlord operating cost (verified) and Henderson–Ioannides’s rental externality (from memory). Evidence for the wedge over the wall: 7.5 percent of renters live in seven-plus-room units and renters hold 8 percent of that stock; 31.8 percent of renters aged 25–45 sit exactly at the cap, a mass point that creates the kinks  $\kappa_H$  smooths.

**Recommendations.** Change (both; medium; blocks).

## H4. Supply and rent

**Agreed.** Replace the instantly reversible schedule  $H_0 P^\eta$  with an inherited stock and a construction margin (ChatGPT:  $I_t \geq 0$ ; Fable:  $I_t = I_0 P_t^\eta$ , the reference paper’s own block, with Kaplan, Mitman and Violante’s construction sector as a second precedent; Coven et al. use the same static form as the model, so the Coven comparison is like-for-like either way). Keep the forward-looking user cost; it is coherent under competitive landlords and a deterministic continuation. Both reviews said the 0.63 elasticity has no derivation from Baum-Snow and Han; the local PDF partly corrects both. Their headline averages are 0.5 for floor space and 0.3 for units across urban neighborhoods (abstract), 0.42 and 0.35 at tract level (pp. 1898–1899), and a metro-aggregate floor-space elasticity of 0.61 to 0.63 appears under their linear-IV specification (p. 1937).

So 0.63 is a metro-level floor-space number from the paper, not a neighborhood or housing-unit elasticity; the deck should cite that page and say which object it is.

**Only ChatGPT.** Define net depreciation consistently with the maintenance owners already pay, so the same physical depreciation is not counted twice. Under successive surprises, use the forecast known at each date, not prices generated by later news. If  $H^s$  means occupied services rather than stock, account for the inherited stock and vacancies.

**Only Fable.** Impact arithmetic: the tax's 17 percent price fall removes about 11 percent of housing within one period under the static rule, while depreciation allows at most 8 percent in four years; this is the "services fall" the September 1 note misread. Long run: with a level supply curve and a population going to zero,  $P \rightarrow 0$  and no per-household stationary limit exists, which is why there is no terminal steady state (ties to D2). Two elasticities coexist in code, 1.75 in the profile chain and 0.63 as a transition override (e1\_profile.py:39; the transition manifest). Live depreciation is 1.1 percent a year (run\_e1\_chain.py:389), not the slide's 2 percent.

**Recommendations.** Change (both; medium). Fable: blocks the transition and  $H_0$  only. ChatGPT: blocks.

## H5. The landlord

**Agreed.** The closure is an outside competitive rental company earning the exogenous return, Greaney et al.'s REIT with the same rent rule (Fable verified). Its stock ownership, financing, tax payments and cash flows must be written; resident welfare excludes nonresident owners unless counted; rent payments and unexpected capital losses do not vanish because expected excess returns are zero. Do not route them through the property-tax rebate (ChatGPT); report them as accounting lines (Fable). Fable verified that the tax base includes the rented stock and that no landlord code exists.

**Only ChatGPT.** Domestic household landlords (Sommer–Sullivan–Verbrugge) are an admissible but non-equivalent alternative closure.

**Only Fable.** In the June 2025 Coven draft on disk, rent is a user-cost no-arbitrage price with the tax inside, so the tax passes to renters there as here; the September 1 note attributes a different rental block to the August 2026 version, which is not on disk, so the difference is unverified either way.

**Recommendations.** Fable: Keep and state (low; no). ChatGPT: Change the documentation and accounts (medium; blocks). Same substance.

## H6. The owner premium $\chi$

**Agreed.** The code applies the premium as  $\chi(h - \bar{h}(m))$  (kernel.py:957–970) and the slide as  $\chi h - \bar{h}(m)$ ; these differ, the latter lowering the owner's physical minimum to  $\bar{h}(m)/\chi$ . Decide which is meant and make code and exposition agree.

**Disagreement: keep or replace.** *ChatGPT:* keep  $\chi$  as a legitimate reduced form; do not call it a missing financial motive, since ownership already provides an asset position and collateral services; do not use it as an unrestricted ownership-fit residual. Verified in the local PDF: Greaney et al. define housing services as  $\chi h + h^r$ , call  $\chi$  the non-pecuniary benefit of ownership, and calibrate  $\chi = 1.0506$  to a homeownership rate of 0.542 (their pp. 10–11 and Table 1); with  $\chi = 0$  no one owns (p. 42). The model's  $\chi = 1.048$  is the same device at the same value, which supports ChatGPT's position; note that the reference form is  $\chi h$ , the slide's, not the code's  $\chi(h - \bar{h})$ . *Fable:* in the model renting and owning the same size cost the same user cost, so  $\chi$  is the only reason to own and stands in for the tax advantages of owning, the rental externality and

the rent-risk hedge; fold it into the intercept of H3’s rental wedge, since the ownership rate identifies only the sum of a premium and a wedge, and the price–rent ratio separates them.

**Recommendations.** Fable: Decide, lean change (medium; blocks). ChatGPT: Keep (low; blocks, the physical-space interpretation).

## Demography, government, closure

### D1. Household, person and dependent accounting

**Agreed.** Write one accounting: births to queue to entrants, survivors to maturation, deaths to estates, units stated, estates reconciled exactly once. The person, household and dependent counts are not linked in the current implementation; person and head aggregates pass their separate checks while the dependent-child link is unresolved. This is a prerequisite for population and intergenerational-allocation claims.

**Disagreement: what the household is.** *ChatGPT:* a unitary household need not contain two adults, and the two-adult base of the scale does not establish a two-person unit. Use an unnormalized household measure with explicit person-stock identities:  $D_{t+1} = D_t + B_t - M_t - \Delta_t^D$  for dependents and  $A_{t+1} = A_t + M_t - \Delta_t^A$  for adults, with household formation a separate mapping from adults to households, not one household per maturing child. *Fable:* three conventions already make it an implicit couple (head-plus-spouse earnings, the two-adult scale base, entrants equal to births over 2.1), so say so; the missing margin is then dissolution by widowhood, which is how most old households become one-person households in large homes (author decision).

**Only Fable.** Entrants arrive twenty years after birth (four waiting slots), as childless renters with PSID 18–24 wealth ratios; the 5 percent leakage inside 2.1 is an external constant while the model’s own orphan leakage is another 5 percent of births; the 2007 stationary normalization forces a replacement population while the childlessness and one-child moments are for the 1960–66 cohort and the timing moments are 2003–06 period statistics.

**Only ChatGPT.** Physical market clearing must use  $h + h^R$ , not  $\chi h + h^R$  (the code does clear in physical units, per Fable’s verifier); orphan transfers are internal movements, not person losses.

**Recommendations.** Fable: Keep and write (low; no). ChatGPT: Change (medium; blocks).

### D2. Population closure

**Agreed.** The prompt describes the wrong branch: the retained branch propagates households endogenously with the births-over-2.1 queue, no imposed masses, no headship law, no migration (both, ChatGPT from M26, Fable from spec. json). The counterfactual must keep the inherited 2023 distribution and pre-2023 birth pipeline and let policy births affect entry only at the later dates. Matching completed fertility to 2.1 in 2007 does not itself establish replacement under the model’s own survival and formation rules.

**Complementary, not conflicting.** *ChatGPT:* choose one demographic interpretation, a literal person model in which survival, entry and formation agree, or an abstract reproductive household with a conversion normalization, and never present converted household counts as persons. *Fable:* the binding problem is the terminal condition. With zero migration and completed fertility below 2.1 there is no positive stationary population ( $B/E \leq 0.87$  at any price), and with level supply no per-household stationary limit either, so the transition solver’s terminal state is a fiction. Two coherent options: a policy-invariant outside inflow (the August 6 closure), or the author’s zero migration with a balanced-contraction terminal condition, which requires H4’s stock-flow supply. The headship law, where used, freezes household formation at 2023 rates.

**Recommendations.** Fable: Decide (low or high). ChatGPT: Change (high). Both: blocks the transition and policy, not the initial calibration.

### D3. PAYGO pensions and the rebate

**Agreed.** Keep. Fixed payroll tax with the benefit adjusting each period is coherent; the equal rebate is Coven’s rule (Fable verified) and defines a particular redistribution experiment rather than an accounting detail. Write the base and recipients:  $T_t \mathcal{H}_t = \tau_t^p P_t (H_t^O + H_t^R)$  when both stocks are taxed at the rate in the user cost (Fable verified that they are). Report capitalization, net tax payments, rebates and pension adjustment separately rather than inferring incidence from age.

**Only ChatGPT.** Any care-pool funding taken from the receipts changes the full-rebate experiment and must be identified.

**Only Fable.** The alternative closure (fixed replacement rate, adjusting tax) would load ageing onto the young; the rebate enters the period budget after the housing choice, not the down-payment test; in a shrinking population pensions fall and that is a downsizing force the model otherwise lacks.

**Recommendations.** Keep (both; none or low).

### D4. Geography

**Agreed.** A national pooled benchmark: remeasure the housing moments nationally under the same sample rules and reconcile prices, supply, earnings and demographic units with that population. The metro alternative needs metro-scoped fertility and demographic evidence and raises migration and sample-support questions. Do not relabel the existing estimates.

**Recommendations.** Change (both; medium; blocks).

## Earnings

### E1. Permanent types and the persistent AR(1)

**Agreed.** Reconcile the income definition, population, taxes, age effects, variances and autocovariances before deleting states; a persistent-only alternative must be refitted to the same evidence; the four-year conversion uses  $\rho^4$  with accumulated innovations. Permanent heterogeneity is not contrary to the mechanism.

**Difference of evidence, not of principle.** *ChatGPT:* the ledger does not establish compatibility or duplication; keep provisionally and remove only if a comparably disciplined persistent-only process reproduces the evidence. *Fable, from code:* the live AR(1) is Floden–Lindé scaled to after-tax units (annual  $\rho = 0.9136$ , innovation variance 0.0426 times  $(1 - 0.181)^2$ , stationary log-variance 0.173) while the fixed effect is the E6b PSID gross-earnings variance 0.393; persistent-plus-fixed is 0.566 in the model against 0.725 in the decomposition the fixed effect came from, in different tax units (0.264 after the same scaling). Not double counting; an incoherent sum. Fix: use the E6b decomposition for all three components with one tax treatment, then run the removal comparison. The types were added on July 27 to reach the old-age wealth tail; the age profile  $e_a$  has no source.

**Recommendations.** Fable: Change (low; blocks). ChatGPT: Keep (medium; blocks, reconcile the evidence). Same procedure; Fable supplies the finding.

## E2. Earnings risk and fertility

**Agreed.** Keep. The channel already exists through continuation values and the option value of waiting; no second uncertainty term. Do not assert that every mean-preserving spread lowers fertility; the object is the difference between having a child and waiting. Fable adds that at four years the AR(1) persistence is 0.70, which mutes the channel.

**Recommendations.** Keep (both; none or low; no).

## Cross-cutting

### X1. Interactions

**Shared.** F3 with F4 (one maturation change may remove the orphan problem); rental segmentation with credit and  $\chi$ ; landlord ownership with taxes and welfare; the scale normalization with  $\psi, \xi, \kappa_1, \kappa_C$ .

**Only ChatGPT.** Keep the  $\psi_0$  root as an initial-calibration restriction but never reapply it in a counterfactual to restore 2.1; that would delete part of the response being studied. Moment-to-parameter labels are not identification.

**Only Fable.** H4 with D2: stock-flow supply is what makes a balanced-contraction long run possible. P1, P3 and E1 meet at the income gradient of fertility (E6b's reversed childlessness gradient is their joint symptom). P5 with  $\beta$  at its cap. D4 with every estimated parameter.

**Sequencing, both orders.** ChatGPT: units and demographic closure; then preference and credit interpretations; then earnings and geographic scope; then identify the revised vector. Fable: F3 and E1 (parameter-level, cheap); then H1 and F2 in one refit; then H4 with D2's terminal condition; then D4 with the final refit.

### X2. The smallest defensible set

**Shared package.** Reconcile dependents, persons and entrants (D1, F3, F4); formalize purchase credit and incumbent debt (H1, H2); specify landlord, estate and fiscal accounts (H5, P5, D3); one empirical population with a compatible earnings process (D4, E1); an inherited housing stock (H4); the shock normalization and the housing-smoothing treatment (F2). Both retain the separate child states, the attempt-and-success structure, the borrowing core and the warm-glow bequest.

**Where the packages differ.** ChatGPT classes concave child benefits, commitment-before-birth timing and richer mortgage instruments as conditional extensions, and calls bounded dependency the largest household-state decision. Fable puts amortization with origination-only borrowing inside the package, names the time cost of children as the largest omission, and lists the items that need explanation rather than change (P2, P4, F1, F5, H5, D1, D3, E2). Both agree that housing adjustment around births does not validate the reverse effect of housing costs on births.

**Author decisions the package leaves open (union of both lists).** P1 linear versus concave; P3 keep  $\xi$  or replace it with a measured penalty; F3 parent-age hazard versus child-stage state; F4 care pool only if F3 is not adopted; F5 unintended births; H1 parsimonious description versus an amortizing origination-only contract; H3 with H6, cap plus premium versus one wedge; D1 the household unit and widowhood; D2 outside inflow versus balanced contraction; P5 estate receiver and a downsizing shock.

### X3. Not on the list

**Only ChatGPT.** (1) Welfare with endogenous fertility: define the population (for example households alive in 2023 and their remaining utilities), report future cohorts separately, and never sum over different numbers of households without explicit weights and a utility-level convention; more births are not a welfare gain by themselves. (2) Omitted nonhousing costs of children may be absorbed into housing and fertility preferences; an externally disciplined goods and time cost is the targeted alternative. (3) The feasible set must enforce rent-or-own; annual flows and four-year utility and discounting conventions must match; the one-birth-per-period restriction must be acknowledged. (4) A discount factor at its cap is a diagnostic of fit or identification, not proof of invalidity.

**Only Fable.** (1) No time cost of children: the cost of a child is a homothetic scale plus an absolute floor, so the model cannot produce the negative income–fertility gradient; both Sommer and Doepke–Kindermann carry the time cost. (2) No household-formation margin; the headship law freezes it. (3) No widowhood or health-driven downsizing. (4) The taste shocks must be declared mean-zero type-I extreme value (no Euler constant in the log-sum, `utils.py:194–198`). (5) Depreciation compounds while the property tax is four times annual (`parameters.py:163–165`). (6)  $\beta R > 1$  at the cap because the old have no expense risk, no return above  $R_b$  and no inheritances. (7) Owner services are  $\chi(h - \bar{h})$  in code,  $\chi h - \bar{h}$  on the slide. (8)  $h_P$  at its upper bound with the per-child slope restricted to zero, so the two rooms moments pull one parameter in opposite directions. (9) One price per room at every size, against hedonic evidence. (10) The fecundity schedule is set in `run_e1_chain.py:380`, invisible from the calibration layer.

**Both, with different weight.** The time cost of children: Fable’s largest omission, ChatGPT’s conditional alternative.

### Combined table

Costs and blocking flags are each reviewer’s own. “Same” means the two recommendations coincide in substance even when the one-word label differs.

| Label               | Fable                                | ChatGPT                  | Agreement             | Cost (F / C)     | Blocks recalibration (F / C) |
|---------------------|--------------------------------------|--------------------------|-----------------------|------------------|------------------------------|
| P1 Utility form     | Decide                               | Keep                     | Differ on action      | Low / Low        | Yes / Yes                    |
| P2 Cross-partial    | Keep (restate on $u$ )               | Change (restate on $u$ ) | Same                  | None / Low       | No / No                      |
| P3 First-birth cost | Decide, lean change                  | Keep                     | Differ                | Low / Low        | Yes / No                     |
| P4 Two child states | Keep                                 | Keep                     | Same                  | None / Low       | No / No                      |
| P5 Bequests         | Change (net estate, receiver)        | Keep (net estate)        | Differ on receiver    | Medium / Low     | Yes / Yes                    |
| F1 Timing           | Keep                                 | Keep                     | Same                  | None / Low       | No / No                      |
| F2 Taste shocks     | Change ( $\kappa_H = 0$ or estimate) | Change (fixed numerical) | Differ on replacement | Low–medium / Low | Yes / Yes                    |

| Label                 | Fable                                  | ChatGPT                                                     | Agreement                 | Cost (F / C)        | Blocks recalibration (F / C) |
|-----------------------|----------------------------------------|-------------------------------------------------------------|---------------------------|---------------------|------------------------------|
| F3 Maturation         | Change (parent-age hazard)             | Change (child-stage state)                                  | Differ on fix             | Low–medium / Medium | Yes / Yes                    |
| F4 Orphans            | Change via F3                          | Change (care pool if needed)                                | Same if F3 adopted        | None / Medium       | Yes / Yes                    |
| F5 Fecundity          | Keep (decide unintended births)        | Keep                                                        | Same                      | Low / Medium        | No / Yes                     |
| H1 Constraint set     | Change (amortization, origination LTV) | Change (description only)                                   | Differ on depth           | Medium / Medium     | Yes / Yes                    |
| H2 Rollover           | Change via H1                          | Change (contract rule)                                      | Same direction            | Low / Medium        | Yes / Yes                    |
| H3 Rental cap         | Change (wedge baseline)                | Change (cap baseline, soft-tail robustness)                 | Differ on baseline        | Medium / Medium     | Yes / Yes                    |
| H4 Supply             | Change (stock-flow)                    | Change (stock-flow)                                         | Same                      | Medium / Medium     | Transition / Yes             |
| H5 Landlord           | Keep and state                         | Change documentation                                        | Same                      | Low / Medium        | No / Yes                     |
| H6 Owner premium      | Decide, lean wedge                     | Keep                                                        | Differ                    | Medium / Low        | Yes / Yes                    |
| D1 Accounting         | Keep and write                         | Change                                                      | Same task, differ on unit | Low / Medium        | No / Yes                     |
| D2 Closure            | Decide                                 | Change                                                      | Complementary             | Low or high / High  | Transition / Yes             |
| D3 Fiscal             | Keep                                   | Keep                                                        | Same                      | None / Low          | No / Yes                     |
| D4 Geography          | Change (national)                      | Change (national)                                           | Same                      | Medium / Medium     | Yes / Yes                    |
| E1 Earnings           | Change (one decomposition)             | Keep (reconcile first)                                      | Same procedure            | Low / Medium        | Yes / Yes                    |
| E2 Risk and fertility | Keep                                   | Keep                                                        | Same                      | None / Low          | No / No                      |
| X1 Interactions       | order: F3, E1; H1, F2; H4, D2; D4      | order: units, closure; preferences, credit; earnings, scope | Differ on order           | n/a                 | n/a                          |
| X2 Minimum set        | 7 changes, 8 explain, 6 decisions      | package above; extensions conditional                       | Large overlap             | n/a                 | n/a                          |

| Label      | Fable                                                 | ChatGPT                                                          | Agreement                  | Cost (F / C) | Blocks<br>recalibration (F<br>/ C) |
|------------|-------------------------------------------------------|------------------------------------------------------------------|----------------------------|--------------|------------------------------------|
| X3 Missing | time cost,<br>formation,<br>widowhood,<br>conventions | welfare<br>population,<br>omitted child<br>costs,<br>conventions | Disjoint, both<br>included | n/a          | n/a                                |

## Appendix. Verification status of claims made by one reviewer only

Fable’s code claims carry file and line in its review. ChatGPT’s claims were checked here where a local source exists:

- ChatGPT: the objective in Scholz, Seshadri and Khitatrakun contains  $e U(c/e)$ , the scale multiplying utility. Verified in docs/reference/scholz\_seshadri\_khitatrakun\_2006\_jpe.pdf, p. 615: “Expected lifetime utility is then  $E[\sum \beta^{j-S} n_j U(c_j/n_j)]$ ”; p. 619:  $n_j = (A_j + 0.7K_j)^{0.7}$ .
- ChatGPT: Greaney et al. use an owner-service premium. Verified in the Zotero PDF (MAXJ699L): services  $\chi h + h^r$ ,  $\chi$  “captures the non-pecuniary benefits of ownership”, calibrated  $\chi = 1.0506$  (pp. 10–11, Table 1).
- ChatGPT: Baum-Snow and Han report about 0.5 for floor space and 0.3 for units. Verified in the Zotero PDF (R5GI43BX), abstract; tract-level 0.42 and 0.35 on pp. 1898–1899. Correction to both reviews: a metro-aggregate floor-space elasticity of 0.61–0.63 appears on p. 1937, so the project’s 0.63 has a page, though not the headline object.
- ChatGPT: Kaplan, Mitman and Violante distinguish origination limits from limits on existing mortgages. Verified, p. 3295: after origination there is no requirement that the outstanding principal stay below  $\lambda_m$  times the home’s value; p. 3290: constraints bind only at origination.
- ChatGPT: Sommer–Sullivan–Verbrugge model household landlords. Not on disk; cited from memory by both.
- Fable: Venti and Wise (2004) on old-age housing equity; Henderson and Ioannides (1983) on the rental externality. Not on disk; from memory.